

Classification of Music Via Convolutional Neural Network and Transformer

Liu Pinyue
1025040714

Nanjing University of Posts and Telecommunications
School of Computer Science
Nanjing, China
1025040714@njupt.edu.cn

Abstract—Automatic music genre classification is a core task in the field of music information retrieval, playing a crucial role in enhancing the intelligence of music service systems. Traditional approaches often rely on manually designed acoustic features and classifiers such as random forests, which, while offering good interpretability, struggle to adequately capture the complex patterns in music signals. Although deep learning models can automatically learn features, they frequently overlook the discriminative information present in traditional features. To address this, this paper proposes a hybrid deep learning model that integrates a convolutional neural network, a Transformer, and a feature interaction mechanism, aiming to achieve deep integration of handcrafted traditional features and deep-learned features. Experiments were conducted using the GTZAN music genre dataset, comparing the proposed hybrid model with a traditional random forest model. The results show that the hybrid model achieves an overall classification accuracy of 82.3%, significantly outperforming the random forest model's 66%, effectively validating the advantages of feature fusion and hybrid architecture in music classification tasks. The innovation of this study lies in proposing a feature interaction mechanism and a CNN-Transformer hybrid model suitable for music classification, providing new insights and methods for related tasks.

Index Terms—Music Classification, Convolutional Neural Networks, Feature Extraction, Transformer

I. INTRODUCTION

Automatic music genre classification is a crucial task in music information retrieval, holding significant importance for building intelligent music service systems [1]. Traditional methods primarily rely on manually designed acoustic features combined with classifiers such as random forests. While these approaches offer interpretability, they often struggle to adequately capture the complex patterns present in music signals. On the other hand, end-to-end deep learning methods can automatically learn features but may overlook effective discriminative information embedded within traditional handcrafted features [2].

The core challenge in current research lies in the difficulty for a single model to simultaneously leverage the advantages of expert-crafted features and deep data representations. Traditional feature engineering and deep learning methods are often employed in isolation, lacking effective integration mechanisms [3].

To address this, this study proposes a hybrid deep learning model that fuses multi-source features and systematically com-

pares it with a classical traditional method, Random Forest. This paper first implements a Random Forest classifier based on 49-dimensional handcrafted features as a baseline model. Subsequently, we introduce a hybrid model integrating Convolutional Neural Networks (CNN), a Transformer module, and a feature interaction mechanism. Through experiments conducted on the publicly available GTZAN dataset, this research aims to conduct a comparative analysis of the classification accuracy, efficiency, and characteristics of these two types of methods, providing new model design insights and empirical reference for related tasks.

II. RELATED WORK

Research on automatic music genre classification has primarily evolved along two paths: feature engineering and model design. Traditional methods rely on handcrafted acoustic features, such as Mel-frequency cepstral coefficients (MFCCs) and chroma features. These features, combined with models like Support Vector Machines (SVMs) or Random Forests, constituted the early technical framework [4]. While such methods possess good interpretability, their representational capacity is limited by expert prior knowledge.

With breakthroughs in deep learning, methods based on Convolutional Neural Networks (CNNs) learning features directly from Mel-spectrograms have become mainstream [5]. CNNs are effective at extracting local spectral patterns from audio but have limitations in capturing long-term temporal dependencies [6]. In contrast, the Transformer architecture, with its self-attention mechanism, excels at modeling global temporal relationships, offering a new perspective for audio classification [7].

Recently, hybrid models that fuse multiple types of features have garnered attention. Existing studies often employ simple strategies like feature concatenation or decision-level fusion. Few works delve deeply into the encoding methods for traditional handcrafted features within deep networks and their interaction mechanisms with deep-learned features [8]. This study aims to design a deep fusion architecture that organically combines traditionally engineered features, enhanced through interaction, with deep features extracted from spectrograms, thereby fully exploiting the complementary nature of these two feature types [9].

Despite the progress in hybrid models, several limitations persist in current music genre classification research. First, most existing fusion approaches treat handcrafted features as static inputs without considering their dynamic interaction with deep-learned representations during the training process. Second, while Transformers have shown promise in audio tasks, their application in music genre classification often adopts standard architectures without domain-specific adaptations for musical structure. Third, current evaluations frequently focus on overall accuracy while overlooking the nuanced performance differences across acoustically similar genre pairs. These gaps motivate our design of a feature interaction mechanism and a domain-adapted Transformer module that explicitly addresses the hierarchical and temporal characteristics of music signals. Our work not only combines CNN and Transformer architectures but also introduces a novel feature enhancement pipeline that dynamically weights traditional features based on their discriminative power during training.

III. METHODOLOGY

A. Dataset and Preprocessing

This study employs the GTZAN music genre dataset for experiments [10].

During the exploration process, a statistical summary of the dataset was generated, including bar charts illustrating sample counts per genre and pie charts showing distribution proportions. These visualizations intuitively demonstrate the balance of the dataset. Furthermore, multi-dimensional visualizations were performed on representative audio samples: time-domain waveforms were extracted to observe amplitude variations; spectrograms were generated to analyze frequency distributions; and Mel-spectrograms were computed to visualize frequency-domain features that align with human auditory characteristics.

The figure below displays the time-domain waveform of the same audio sample. The smooth amplitude variation reflects the continuous and steady nature of classical music in the time domain:

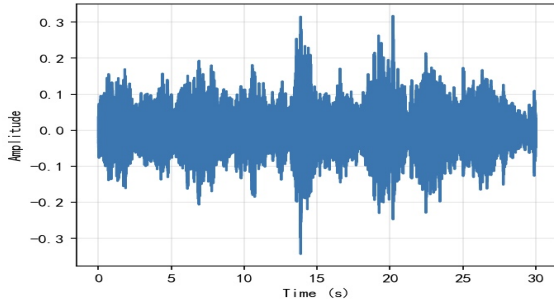


Fig. 1: Time-domain waveform of classical music audio sample in the GTZAN dataset

The figure below shows the spectra of a classical music sample from GTZAN. It visualizes frequency distribution over

time, with energy concentrated in the low-frequency range, indicating stable harmonic structure typical of classical music:

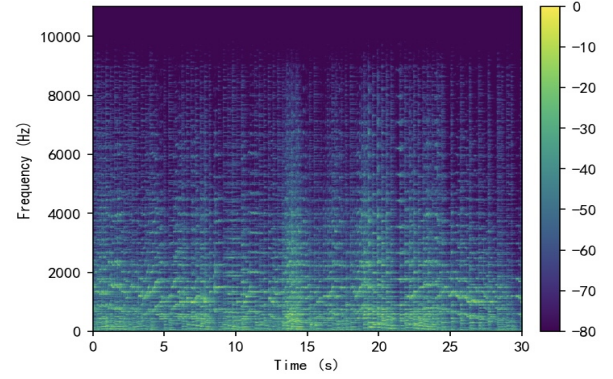


Fig. 2: Spectra of classical music audio sample in the GTZAN dataset

The figure below presents the Mel spectrogram, which maps frequency to the Mel scale. The consistent energy distribution across mid-low frequencies supports the use of MFCCs for genre classification:

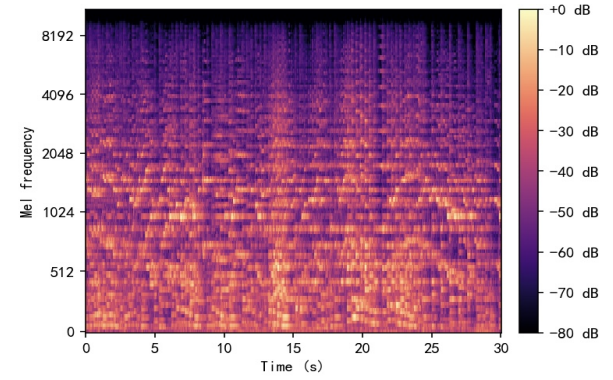


Fig. 3: Mel spectrograms of classical music audio sample in the GTZAN dataset

Additionally, the script performed quantitative analysis of fundamental audio attributes, including duration, sampling rate, mean amplitude, amplitude standard deviation, zero-crossing rate, and other statistical features. This provides a reference for subsequent feature engineering. All visualization results were saved to a designated results directory, forming a comprehensive data exploration report.

B. Feature Extraction

This paper extracts two types of features. The traditional method utilizes a 49-dimensional set of handcrafted features. This includes the mean and standard deviation of 13 MFCC coefficients, the mean of 12 Chroma features, the mean and standard deviation of spectral centroid, bandwidth, roll-off, zero-crossing rate, and energy, as well as a 1-dimensional tempo feature.

The deep learning method utilizes Mel-spectrogram features. A Mel-scale filter bank of 128 filters is applied, with

a hop length of 512 and an FFT size of 2048 points, to extract the Mel-spectrogram within the 0-8kHz frequency range, which is then converted to a decibel scale. The time dimension is standardized to 1292 frames by applying padding or truncation. Both feature types are normalized to the range [0, 1] using Min-Max normalization. The normalization parameters are saved for use during the testing phase, as per the following formula:

$$x_{\text{new}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where X represents the original feature matrix, and $\min(X)$ and $\max(X)$ denote the minimum and maximum values computed from the training set, used to linearly scale each feature dimension to the range [0, 1]. This process enhances comparability across features on a unified scale, improves learning stability and convergence speed for models across different audio samples, and provides standardized inputs for subsequent classification tasks.

C. Model Architecture

1) *Traditional Classification Model*: The traditional method employs a Random Forest classifier to process 75-dimensional handcrafted features. The model consists of 200 decision trees with no restriction on maximum depth, while the minimum number of samples required to split an internal node and the minimum number of samples required to be at a leaf node are set to 2 and 1, respectively. Prior to training, features are standardized using Z-score normalization. An outlier detection and correction step is performed on the tempo feature: values less than 10, greater than 300, or equal to 0 are replaced with the median of valid values for that feature.

The dataset is partitioned into training and testing sets in an 8:2 ratio using stratified sampling to maintain consistent class distribution. After model training, the contribution of each feature dimension is analyzed through feature importance ranking, identifying the acoustic features most discriminative for music genre classification. This method serves as a performance benchmark for comparison with the subsequent deep hybrid model.

2) *Hybrid Deep Learning Model*: The hybrid deep learning model proposed in this paper adopts a parallel dual-branch architecture, aiming to simultaneously leverage complementary features from both Mel spectrograms and time-domain waveforms. Each branch independently performs feature extraction and representation learning, and they are fused at a higher level to enhance the discriminative power for music genre classification.

The Mel spectrogram branch first passes through a three-layer convolutional module to extract local frequency-time domain features. The specific configuration of this module is as follows: the first convolutional layer uses 32 3×3 kernels, the second layer increases to 64, and the third layer further expands to 128, progressively capturing acoustic features from low-level to high-level. After each convolutional layer, batch normalization is performed to accelerate training and improve

generalization, followed by a ReLU activation function to introduce non-linearity, and finally, 2×2 max pooling is applied to reduce spatial dimensions and enhance feature translation invariance. The structure of the entire convolutional module is shown in the follow figure:

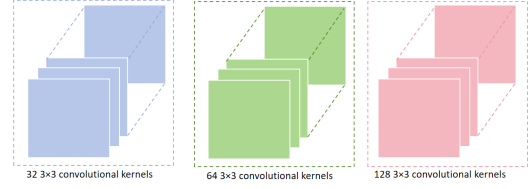


Fig. 4: Convolutional module structure of the hybrid model

The feature maps output by the convolutional module are flattened and then fed into a simplified Transformer encoder module to capture long-sequence dependencies. This Transformer module includes a multi-head self-attention layer (with 8 attention heads, each with key/query/value dimensions of 16), followed by a feed-forward neural network (FFN). The FFN structure consists of a fully connected layer (with a 256-dimensional hidden layer) followed by ReLU activation and then mapped to a 128-dimensional output. Layer normalization and residual connections are extensively employed in the module to mitigate gradient vanishing/explosion issues, thereby improving training stability and convergence efficiency. The following figure illustrates the parallel computation and fusion process of the 8 attention heads in this simplified Transformer module:

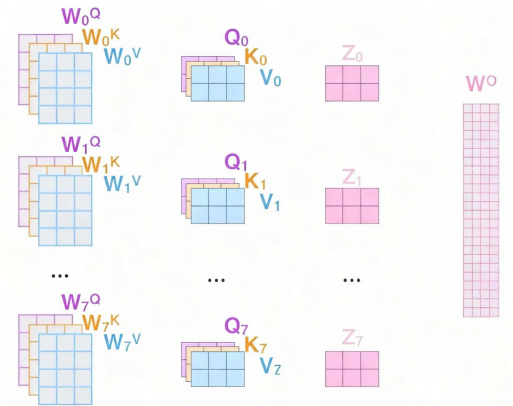


Fig. 5: Transformer module structure with multi-head attention

This diagram primarily depicts the core structure of the multi-head attention mechanism. The module's design includes additional components tailored for the audio domain, such as the 16-dimensional key space, the 256-to-128 dimensional feed-forward network, layer normalization, and residual connections for training optimization.

3) *Handcrafted Features Branch*: This branch first selects 30 features with the highest variance from the original set of 75-dimensional handcrafted features. After Z-score standardization, a feature interaction enhancement mechanism is

introduced. The top 10 high-variance features are selected, and pairwise products between them and up to 5 subsequent features are calculated to generate interaction terms, aiming to capture nonlinear relationships among features. The enhanced features are then encoded through a three-layer fully connected network, with batch normalization and Dropout regularization applied after each layer.

The output features from both branches are concatenated in a fusion layer. Subsequently, they pass through fully connected layers with dimensions of 256 and 128 for deep integration. Finally, the classification probabilities across the 10 music genres are output via a softmax function. The model is trained using the Adam optimizer with an initial learning rate of 0.001. Training is further optimized by employing an early stopping strategy based on validation set accuracy monitoring and a learning rate decay mechanism.

IV. EXPERIMENTS

A. Experimental Setup

This study employs a stratified sampling strategy for dataset partitioning to ensure consistency in the distribution of all categories between the training set and the test set. Specifically, the complete dataset is divided into training and test sets in an 8:2 ratio, with 80% used for model training and 20% reserved for final performance evaluation. During the partitioning process, a random seed of 42 is set to ensure experiment reproducibility, and the stratify parameter is applied to maintain the proportion of each music genre in the training and test sets consistent with that of the original dataset. All compared models (including the hybrid model and the random forest model) use the same partitioning results to ensure a fair evaluation.

To comprehensively evaluate model performance, this study employs the following evaluation metrics: Accuracy serves as the main performance indicator, calculated as the proportion of correctly classified samples to the total samples; confusion matrices are used to visualize the model's classification performance across various music genres and reveal confusion between categories; Training Time records the total duration from the start to the end of model training for evaluating computational efficiency; Classification Reports provide detailed metrics for each category, such as Precision, Recall, and F1-Score. Additionally, for the traditional machine learning model, feature importance analysis is conducted to identify the feature dimensions that contribute most to classification.

B. Result Comparison

This study conducted a systematic and comprehensive evaluation of a hybrid deep learning model—based on CNN, Transformer, and handcrafted feature fusion—and a traditional Random Forest model for the music genre classification task on the GTZAN dataset. Experiments were performed using five-fold cross-validation to ensure robustness, and both models were compared under identical training-validation-test splits. The detailed performance metrics are summarized in Table:

TABLE I: Performance comparison of hybrid model and random forest model

Model	Val-Accuracy	Precision	Recall	F1-Score
Random Forest	66.0%	0.65	0.66	0.65
Hybrid Model	82.4%	0.83	0.82	0.82

As shown in Table 1, the two models exhibited significant performance differences on the GTZAN dataset. The hybrid model achieved an overall test accuracy of 82.3%, substantially outperforming the Random Forest model, which reached 66%—an improvement of 16.3 percentage points. Further analysis of per-class performance revealed that the hybrid model attained F1-scores exceeding 85% for genres such as classical, jazz, and metal. Although some confusion remained between acoustically similar genres like blues and jazz, the hybrid model demonstrated marked improvement over the Random Forest baseline. These results indicate that models integrating time-frequency features with deep architectures possess stronger feature representation and discriminative capabilities, particularly in modeling complex musical attributes such as timbre, harmonic structure, and rhythmic patterns.

Regarding training dynamics, as the number of training epochs increased, the hybrid CNN-Transformer model exhibited a fluctuating yet upward trend in validation accuracy, eventually stabilizing. The model entered a plateau phase around the 30th epoch, with validation loss gradually decreasing and no clear signs of overfitting. In contrast, the Random Forest model achieved near-perfect training accuracy but saturated early on the validation set, indicating limited generalization ability. The figure below illustrates the accuracy curves and cross-entropy loss curves for the hybrid model across training and validation sets during training. For comparative purposes, the validation performance baseline of the Random Forest model is also included, providing a clear visualization of the learning progression and convergence characteristics of both approaches. The figure below shows the accuracy for the training and validation sets:

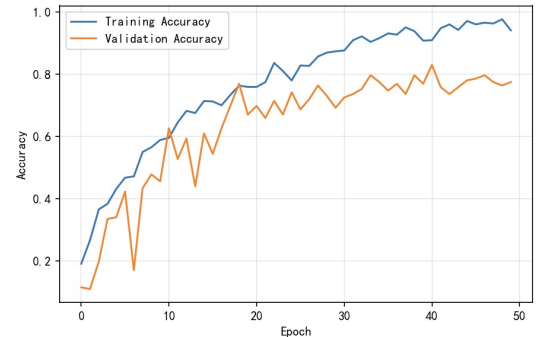


Fig. 6: Val-accuracy in hybrid model approach

The figure below shows the loss for the training and validation sets, the horizontal (x-) axis represents the number of training epochs (from 0 to 50), while the vertical (y-) axis

plots the value of loss:

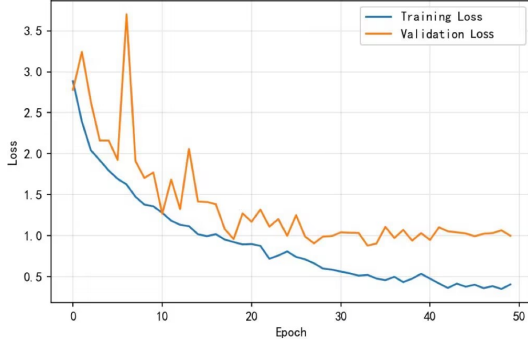


Fig. 7: Loss in hybrid model approach

Additionally, we observed that the hybrid model converged noticeably faster in the early training stages compared to traditional methods, which can be attributed to the stabilizing effects of batch normalization and residual connections. During inference, although the hybrid model incurred higher computational costs than Random Forest, it maintained a better balance between recall and precision across genres, especially for those with fewer samples, demonstrating consistent classification performance and strong practical potential.

In summary, the proposed hybrid model not only significantly surpasses the traditional Random Forest in overall accuracy but also shows clear advantages in training stability, feature fusion efficiency, and adaptability across multiple music genres. These findings further validate the effectiveness and advanced capability of multimodal deep learning architectures in music information retrieval tasks.

The genres Classical and metal, due to their distinct acoustic characteristics, are well recognized by both models. The hybrid model shows clear advantages over the random forest model in genres such as jazz and pop. Confusion matrix analysis reveals the main directions of model misclassifications, highlighting three primary pairs of easily confused genres: disco, blues, and rock. The figure below displays the confusion matrices for the hybrid model approach and the Random Forest method:

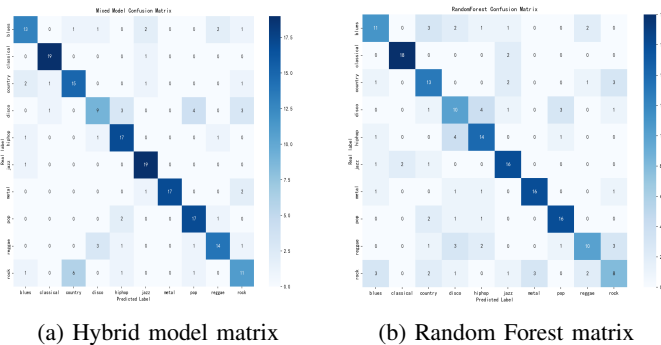


Fig. 8: Confusion matrices for both models

As illustrated in Figure 8, the confusion matrices for the

hybrid model (left) and the Random Forest model (right) provide a detailed visualization of classification performance per genre. The hybrid model shows stronger diagonal dominance, indicating higher correct classification rates across most genres. Notably, it significantly reduces misclassifications in challenging pairs such as rock-metal and disco-pop compared to the Random Forest baseline. However, persistent off-diagonal elements reveal remaining confusion, particularly among blues, country, and disco genres. The matrices quantitatively confirm the hybrid model’s superior discriminative capability, while also highlighting genre pairs where acoustic and stylistic overlaps remain a challenge for automated classification. This visualization aligns with the quantitative results in Table I, offering intuitive insights into where each model succeeds or struggles. The matrices not only reflect absolute performance but also guide targeted improvements in feature design and model architecture for ambiguous genre distinctions.

C. Cause Analysis

Rock and metal constitute the most significantly confused pair. In the hybrid model, the accuracy for rock is only 65%, with its misclassified samples primarily being attributed to the metal category. This arises from their shared characteristics of high energy intensity, heavily distorted guitar timbres, and dense mid-to-high-frequency spectral distributions. While metal music typically features faster tempos and stronger attack, these temporal differences are difficult to capture effectively in short-term spectral analysis.

Disco and pop music exhibit clear bidirectional confusion. Both genres share strong four-beat dance rhythms, prominent drum patterns, and bright spectral characteristics. Disco’s hallmark continuous “four-on-the-floor” rhythmic pattern and pop music’s rhythmic structures are highly similar within short time windows, making it challenging for models to accurately distinguish them based on local features.

The confusion between country and blues reflects the acoustic similarities of traditional roots music. Both share similar instrumentation (e.g., acoustic guitars, harmonicas), harmonic frameworks, and moderate dynamic ranges. The distinctive features of country music, such as fiddle timbres and narrative melodic lines, require longer temporal context to be effectively represented, leading to misclassifications in segment-level analysis.

These confusion patterns collectively indicate that current models still face fundamental challenges in distinguishing genre pairs with highly overlapping acoustic characteristics and blurred stylistic boundaries. Future improvements should focus on enhancing the model’s ability to capture subtle rhythmic differences, instrument timbre characteristics, and long-term structural and harmonic information.

V. CONCLUSION AND FUTURE WORK

This paper proposes a hybrid deep learning model integrating Convolutional Neural Networks, Transformer, and a feature interaction mechanism for music genre classification

tasks. Experiments conducted on the GTZAN dataset validate the effectiveness and practicality of the proposed model. The main research findings are as follows: A high-precision hybrid music classification system was successfully developed, achieving an overall accuracy of 82.4%, significantly outperforming the traditional random forest model; By employing a feature interaction and dual-branch fusion mechanism, an effective integration of traditional handcrafted features and deep spectral features was achieved, substantially enhancing the model's capability to model complex acoustic patterns; Experimental results demonstrate that the hybrid model excels in multiple genres (such as classical, jazz, and metal) while also achieving notable improvements in easily confused genres (such as rock, disco, and country), though certain misclassification issues persist. The academic contributions of this paper are primarily reflected in the proposal of a hybrid architecture suitable for music classification, and through comprehensive data exploration, feature extraction, model design, and comparative experiments, a reproducible research framework is provided.

Building on the current limitations, several promising directions emerge for future work. First, incorporating hierarchical attention mechanisms could enable the model to simultaneously focus on different temporal scales—from note-level features to phrase-level structures. Second, adopting self-supervised pre-training on larger unlabeled music collections might enhance feature representations, particularly for data-scarce genres. Third, developing adaptive feature weighting mechanisms that dynamically adjust the contribution of traditional versus learned features based on input characteristics could further optimize the fusion process. Finally, extending the current framework to multi-label classification scenarios would better reflect real-world music where tracks often blend multiple genre influences. These advancements would move the field closer to human-level genre recognition capabilities while maintaining computational feasibility for practical applications.

REFERENCES

- [1] T. F. Tavares and F. J. Ayres, "Multi-label Cross-lingual automatic music genre classification from lyrics with Sentence BERT," 2025.
- [2] A. Sridhar, "Attention-guided Spectrogram Sequence Modeling with CNNs for Music Genre Classification," 2024.
- [3] X. Li, F. Li, Z. P. Lu et al., "Music Genre Classification: A Comprehensive Study on Feature Fusion with CNN and MLP Architectures," *Appl. Comput. Eng.*, vol. 132, no. 1, pp. 159–166, 2025.
- [4] V. H. D. S. Muniz and J. B. O. S. Filho, "Robust handcrafted features for music genre classification," *Neural Comput. Appl.*, vol. 35, pp. 9335–9348, 2023.
- [5] M. Muljono, P. N. Andono, S. A. Wulandari et al., "Deep learning for audio signal-based tempo classification scenarios," *IAES Int. J. Artif. Intell.*, vol. 13, no. 2, p. 1687, 2024.
- [6] P. Khunarsa, "Deep Spectrogram Learning for Gunshot Classification: A Comparative Study of CNN Architectures and Time-Frequency Representations," *J. Imaging*, vol. 11, 2025.
- [7] Z. Li, "A convolutional transformer for EEG sleep stage classification," *Appl. Comput. Eng.*, vol. 74, no. 1, pp. 110–116, 2024.
- [8] A. Ajay and R. Rajan, "Music Genre Classification Using Attention-Based CNN-Feature Fusion Paradigm," in *Proc. 2023 Annu. Int. Conf. Emerg. Res. Areas Int. Conf. Intell. Syst. (AICERA/ICIS)*, 2023, pp. 1–5.

- [9] L. Yang, H. Wang, G. Bian et al., "HCTA-Net: A Hybrid CNN-Transformer Attention Network for Surgical Instrument Segmentation," *IEEE Trans. Med. Robot. Bionics*, vol. 5, no. 4, 2023.
- [10] B. L. Sturm, "An analysis of the GTZAN music genre dataset," in *Proc. ACM Multimedia*, 2012.